



Homeostatic dynamics, hysteresis and synchronization in a low-dimensional model of burst suppression

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Abstract Burst suppression, a pattern of the electroencephalogram characterized by quasi-periodic alternation of high-voltage activity (burst) and isoelectric silence (suppression), is typically associated with states of unconsciousness, such as in deep general anesthesia and certain etiologies of coma. Recent computational models for burst suppression have attributed the slow (up to tens of seconds) time-scale of burst termination and re-initiation to cycling in supportive physiological process, such as cerebral metabolism. That is, activity-dependent substrate (‘energy’) depletion during bursts, followed by substrate recovery during suppression. Such a model falls into the category of a fast-slow dynamical system, commonly used to describe neuronal bursting more generally. Here, following this basic paradigm, we develop a low dimensional mean field model for burst suppression that adds several new features and capabilities to previous models. Most notably, this new model includes explicit homeostatic interactions wherein the rates of substrate recovery are tied to neuronal activity in a supply demand loop, creating a physiologically consistent, reciprocal interaction between the neural and substrate processes. We develop formal analysis

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of the model dynamics, showing, in particular, the capability of the model to produce burst-like activity as a consequence of neuronal downregulation only, without any direct perturbation to the substrate dynamics. Further, we use a synchronization analysis to contrast different mechanisms for spatially local versus global bursting. The analysis performed generates characterizations that are consistent with experimental observations of spatiotemporal features such as burst onset, duration, and spatial organization and, moreover, generates predictions regarding the presence of bistability and hysteresis in the underlying system. Thus, the model provides new dynamical insight into the mechanisms of burst suppression and, moreover, a tractable platform for more detailed future characterizations.

Keywords Burst suppression · Metabolism · Mean field model · Homeostasis · Hysteresis · Burst synchronization

Mathematics Subject Classification 92C20 · 34C15

1 Introduction

Burst suppression is an electroencephalogram (EEG) pattern associated with neurophysiological conditions in which the brain is substantially inactivated, including pharmacologically-induced coma (Brown et al. 2010; Clark and Rosner 1973; Amzica 2009) and coma due to brain injury (Akrawi et al. 1996; Young 2000; Murdoch and Hall 1990; Hall and Murdoch 1990; Amzica 2009). In burst suppression, the EEG alternates between periods of high-voltage activity (bursts) and flatline (suppression) (Amzica 2009; Brenner 1985) (see Fig. 1a). The commonality of burst suppression to multiple etiologies has made it of fundamental interest in basic (Steriade et al. 1994; Amzica and Kroeger 2011; Kroeger and Amzica 2007; Ferron et al. 2009) and clinical neuroscience (Ching and Brown 2014; Amzica and Kroeger 2011).

To account for this commonality, dynamical systems-based models have been recently developed for burst suppression that implicate neuronal metabolism as central to the underlying mechanism (Ching et al. 2012; Westover et al. 2013). In these models, the slow time scales associated with burst and suppression alternation are parametrically modulated through the actions of an adenosine-triphosphate (ATP) gated potassium channel in individual neurons. Termination of each burst is associated with depletion of ATP, the energetic substrate for action potential generation, due to attenuated metabolic recovery. During suppression, cessation of firing allows for ATP to recover until a critical threshold for spiking activity is reached, at which point a new burst is initiated. This mechanism is based on the observation that the etiologies of burst suppression are generally associated with significant reductions in cerebral metabolic rate of oxygen (Lin et al. 2010; Oshima et al. 2002). In a separate study, a model for burst suppression was recently developed using a mean field modeling framework (Liley and Walsh 2013). There, the slow alternations of burst and suppression are attributed to other physiologic processes that gate neuronal activity, such as synaptic neurotransmitter depletion and recovery.

In terms of dynamics, these previous models are, in essence, separated time-scale systems of the form:

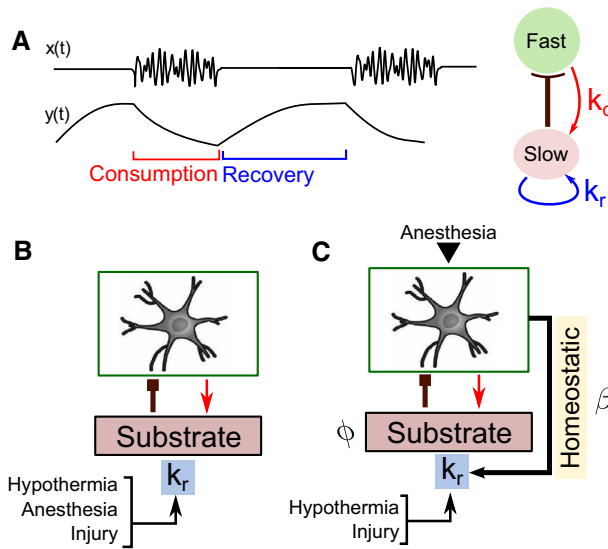


Fig. 1 Model scheme. **a** In burst suppression, neuronal activity ($x(t)$) follows a pattern in which bursts of high voltage activity alternate with periods of suppression (see, e.g., Lewis et al. 2013; Ching et al. 2012; Amzica 2009). Recent modeling efforts (Liley and Walsh 2013; Ching et al. 2012) have attributed this phenomenon to a slower modulating physiologic variable ($y(t)$) that is depleted during bursts and recovers during suppression, resulting in a fast-slow dynamical in the form (1) and (2). The time scales of burst and suppression are dictated by rates of substrate consumption and recovery, k_c and k_r , respectively. **b** In previous models, the etiologies of burst suppression putatively downregulated the substrate dynamics directly, e.g., via k_r . **c** We develop a model that includes homeostatic interactions, wherein the rate of substrate recovery is dependent on neuronal activity in a supply-demand fashion. Such dynamics enable burst suppression to emerge due to neuronal downregulation only, which, in particular, is consistent with the primary actions of anesthetic drugs

$$\dot{x} = f_{\alpha_n}(x, y), \quad (1)$$

$$\dot{y} = \nu g_{\alpha_m}(x, y), \quad (2)$$

where x is the ‘neural’ variable of a fast system (1) and y is the ‘modulation’ variable of a slower process (2) (e.g., ATP or some other supportive substrate; see Fig. 1a for schematic). Such resource-dependent neural models are an emerging area of interest in the context of understanding how the brain maintains homeostasis (Roberts et al. 2014). Here, the parameter ν governs the time scale separation between the fast and slow subsystems. The slow dynamics are parametrized by α_m , which, for (Ching et al. 2012), corresponds to the rate of substrate recovery (k_r in Fig. 1a). Depending on the parameter α_m , the slow subsystem may exhibit a relatively constant steady state (thus supporting continuous activity in x), or periodic cycling (i.e., burst and suppression of x). The period (i.e., suppression epoch duration) grows as α_m is progressively depressed. More generally, such a fast-slow description is common in spiking neuron models that describe burst phenomena at the cellular level (Rinzel 1987; Izhikevich 2007).

The models in Ching et al. (2012) and Liley and Walsh (2013) are successful insofar as they suggest basic mechanisms for burst suppression that are not specific to any one etiology. They also account for characteristic electrophysiological features

of burst suppression such as very slow alternation between burst and flatline and a parametric lengthening of suppression-burst periodicity with respect to progressive brain inactivation. Despite these successes, several key questions remain open. First, these prior models lack homeostatic, autoregulatory interactions between the fast and slow subsystems. That is, α_m is strictly an extrinsically specified parameter and does not depend on any of the dynamical variables. Thus, all etiologies of burst suppression are modeled as perturbations to α_m (Fig. 1b). Such a formulation is appropriate in certain contexts, for example direct metabolic perturbations such as hypothermia or ischemia. In contrast, while anesthetic drugs do decrease cerebral metabolism, this reduction is largely a consequence of a drug-induced reduction in neuronal activity (Morita et al. 1977; Karlsson et al. 1990). That is, α_m is, in general, time-varying and, moreover, homeostatically coupled to neuronal activity (Fig. 1c).

Moreover, recent in vivo characterizations of burst suppression reveal systematic changes in spatiotemporal features such as burst amplitude, duty-cycle, and synchronization (Kenny et al. 2014; Westover et al. 2015) that were not previously modeled. In the case of synchronization in particular, it has been shown that bursts can reside within an isolated cortical region while suppression continues in neighboring regions (Lewis et al. 2013), in contrast to classical descriptions of burst suppression as a whole-brain phenomenon (Clark and Rosner 1973; Amzica 2009).

Here, we develop a modeling paradigm for burst suppression that preserves the essential fast-slow dynamics discussed previously; but that enriches the mechanistic characterization toward closing these gaps. Our contributions are twofold: first, we construct a fast-slow neural mass model based on the neural and metabolic dynamics of Ching et al. (2012) by augmenting a Wilson–Cowan model with a generalized, metabolic slow subsystem, ensuring homeostatic coupling between the neural and metabolic processes (Fig. 1c). This model refines and elaborates on our earlier concepts for a fully coupled neural-metabolic model (Liu and Ching 2014), wherein the metabolic dynamics are closely tied to the neural dynamics. A similar notion has also been recently developed in Bojak et al. (2015) in the context of neurotransmitter cycling, which pursued a homeostatically coupled elaboration of the model of Liley and Walsh (2013), however, with several key differences from the present work (see also Discussion). Second, we analyze the intrinsic dynamics within and between coupled mass models, providing a formal characterization of burst dynamics and their synchronization across space. In the process, we explore the dynamical underpinnings of features such as the dichotomy between local and global bursting (Lewis et al. 2013), the morphology of bursts/suppression and the presence of hysteresis in the induction of (and emergence from) burst suppression with respect to downregulation in the neuronal dynamics.

2 Model formulation

Expanding on Liu and Ching (2014), we develop a low-dimensional fast-slow model for neuronal-substrate dynamics in burst suppression. The model consists of fast dynamics described by Wilson–Cowan equations augmented with generalized, non-specific slow dynamics. The fast and slow systems are homeostatically coupled.

2.1 Cortical mean field model

Since we are not investigating a particular cortical dynamical regime (e.g., frequency of oscillation) per se, we use a simple mean-field description of the Wilson–Cowan type (Wilson and Cowan 1972, 1973) for the fast neuronal dynamics. To the standard Wilson–Cowan equations we add a supportive modulating input $\phi_j(t)$, resulting in:

$$\dot{e}_j = \omega_e \left(-e_j + (k_e - r_e e_j) \mathcal{F}[c_1 e_j - c_2 i_j + \sum_{k \in \mathbb{N}_j} k_j^{f_e} e_k + P + \phi_j(t)] \right) + W_j^e(t), \quad (3)$$

$$\dot{i}_j = \omega_i \left(-i_j + (k_i - r_i i_j) \mathcal{F}[c_3 e_j - c_4 i_j + \sum_{k \in \mathbb{N}_j} k_j^{f_i} e_k + Q] \right) + W_j^i(t). \quad (4)$$

Here, e_j and i_j represent the overall activity in the excitatory and inhibitory populations in a 1–3 mm diameter column of cortical tissue. The parameters c_1 , c_2 , c_3 and c_4 determine intra-column connectivity. ω is the time constant. Coupling between columns is dictated by the parameters $k_j^{f_e}$, $k_j^{f_i}$. Here, $\mathbb{N}_j$ is the index set of the columns from which column j receives afferent excitation. The constants P and Q determine the level of background excitation present in the system. Depending on these values, an isolated column may exhibit either a stable fixed point or a stable limit cycle. As in Wilson and Cowan (1972, 1973), the function $\mathcal{F}$ is the logistic sigmoid

$$\mathcal{F}(x) = \frac{1}{1 + \exp[-a(x - \theta)]} - \frac{1}{1 + \exp(a\theta)}, \quad (5)$$

where a , θ determine the slope and tuning, respectively, of the sigmoid. The noise terms $W_j^e(t)$, $W_j^i(t)$ are included to model weak stochastic input from afferent populations or random exogenous stimuli.

Our model development centers on the variable $\phi_j(t)$, a gating process that supports the fast system as a function of substrate, which may, for instance, reflect the conductance of ATP-gated ion channels (e.g., Ching et al. 2012; Khakh 2001; MacDermott et al. 1999) or efficacy of ATP-dependent ion pumps (Harris et al. 2012). Here, we formulate ϕ_j as a nonspecific, dimensionless metabolic variable. It is modeled abstractly, as an additive term within the sigmoid $\mathcal{F}$, with a slight abuse of notation related to implicit transformation of units enabling addition with the other quantities in this sum. Here, we choose to model ϕ as impinging exclusively on the fast excitatory population.

When $\phi_j(t)$ is sufficiently large, the neuronal population can sustain high voltage oscillatory activity. When it is low, the system will produce a relatively low voltage quiescent steady state (dependent on the background parameter P in Eq. (3)). Burst suppression emerges as a consequence of ϕ_j cyclically modulating the fast system.

It is important to note that the variables e and i in the Wilson–Cowan model represent neuronal firing rates, and can not be directly interpreted as field-potential or EEG activity. However, a strong correlation between firing rates and EEG has been empirically characterized (Allers et al. 2002; Hughes et al. 2011; Hiltunen et al. 2014) and several one-to-one transformations between the two have been proposed (Whittingstall and Logothetis 2009; Mazzoni et al. 2010; Telenczuk et al. 2015; Robinson et al. 2004). Here, we treat the model fast variables as a surrogate for EEG, presuming that bursts and suppression in field potential are, fundamentally, reflections of similar phenomena that are occurring at the level of firing rates (Steriade et al. 1994; Amzica and Kroeger 2011).

2.2 Slow metabolic substrate model with homeostatic coupling

We proceed to introduce a slow system to describe the evolution of $\phi_j(t)$ as a function of neuronal activity. Specifically,

$$\dot{\phi}_j = -\mu_1 \phi_j + \left(\frac{\mu_2}{1 + \exp[-\kappa_\phi(M_j - \eta)]} \right) \quad (6)$$

where $1/\mu_1$, μ_2 are the time scales of the slow process ϕ_j and M_j is the substrate that is regenerated and consumed according to:

$$\dot{M}_j = g_r(e_j) - g_c(e_j). \quad (7)$$

Thus, M_j gates the modulating variable ϕ_j . The consumption/regeneration of M_j by e_j is based on observations that signaling processes (e.g., action potentials) account for most of the total energy consumption in the brain (Howarth et al. 2012; Alle et al. 2009). To again draw a parallel to the mechanism in Ching et al. (2012), M_j would model extracellular ATP (and the dynamics of Sodium-ATP exchange during production of action potentials).

We proceed to define the regeneration and consumption functions $g_c(e_j)$ and $g_r(e_j)$ in a manner that is consistent with these previously characterized biophysics. Specifically, we formulate these functions with sigmoidal shapes typical of models of substrate depletion and consumption (Tabak and Rinzel 2005) and neuronal bursting more generally (Bressloff 2012; Morris and Lecar 1981):

$$\begin{aligned} g_r &= k_r \beta_j, \\ g_c &= k_c \left(\frac{e_j^2}{0.01 + e_j^2} \right), \end{aligned} \quad (8)$$

where k_r , k_c are positive regeneration and consumption constants. The specification of g_c follows from observations that the total energy consumed by neuronal activity has an approximately sigmoidal dependency on spiking rate (Lennie 2003; Attwell and Laughlin 2001). The specific functional form in (8) is chosen as representative of a broader class of possible sigmoidal functions.

The key homeostatic mechanism arises through the variable β_j , the time-evolving recovery rate, dynamically specified as

$$\dot{\beta}_j = -v_1\beta_j + \left(\frac{v_2}{1 + \exp[-\kappa_\beta(e_j - \zeta)]} \right), \quad (9)$$

where $1/v_1$, $1/v_2$ are the characteristic times scales of the slow recovery process. Thus, the dependency of β_j on e_j in (9) implies that changes in excitatory neuronal activity result in compensatory changes in the rate of metabolic recovery, forming a supply-demand homeostatic loop. This dependency is consistent with empirical characterizations of tight coupling between energy demand and supply in the brain, reflected by glucose and oxygen delivery from the cerebral vasculature (see, e.g. Belanger et al. 2011). For example, increased neural demand results in increased supply, wherein changes in neural activity are invariably accompanied by changes in local blood flow and glucose utilization (Takano et al. 2006; Gusnard and Raichle 2001). Thus, in our model, if e_j is downregulated, β_j will decrease, and vice versa. Note that, since ϕ_j modulates neuronal activity, we can use ϕ_j as an approximation for a temporally filtered version of the neuronal activity e_j .

Thus, our specification for ϕ_j , and specifically its homeostatic dependency on neural activity, follows from metabolic arguments. In principle, other types of slow activity could be similarly formulated by adapting the physiological variables and functional forms. For instance, cycling of synaptic neurotransmitter has been modeled in a similar capacity in Liley and Walsh (2013) and Bojak et al. (2015) (see also Discussion).

Together, the system (3)–(9) comprises a five dimensional mean-field model for neural and substrate dynamics. Table 1 summarizes the model variables, their physical interpretation, and the typical values used in our subsequent analysis. We note that the values chosen for the fast subsystem are taken directly from the original formulation of the Wilson–Cowan model in (Wilson and Cowan 1972, 1973) and, as such, are dimensionless. For the remaining parameters, we assume a time scale in milliseconds (as per units in Table 1), though we note that, in general, the time scale of this model can be made arbitrary.

3 Analysis of dynamics in isolated columns

We proceed to characterize the dynamics of a single column, first highlighting consistency with previous models, then demonstrating its ability to generate burst-like dynamics due to downregulation of only the neural subsystem.

3.1 Model produces slow bursting due to lowered substrate recovery rate

We first establish the capacity of the model to produce burst-like activity through direct, extrinsic downregulation of the basal substrate recovery rate parameter k_r . Such a manipulation is analogous to the downregulation of the ATP production rate J_{ATP} in Ching et al. (2012). Figure 2 shows the system output at steady state for several choices of the rate parameters k_c and k_r . As anticipated, reduction of k_r (relative to

Table 1 Parameterization for isolated column/network analysis

Parameter	Definition	Value (units) (isolated)	Value (units) (network)
k_e, k_i	The maximal values of the excitatory, inhibitory response functions	1, 1	1, 1
r_e, r_i	The absolute refractory period of the excitatory, inhibitory subpopulation	1, 1	1, 1
ω_e, ω_i	Wilson Cowan column time constants	0.5, 0.5 (ms^{-1})	0.5, 0.5 (ms^{-1})
k_j^{fe}, k_j^{fi}	Synaptic coupling strength between columns	0, 0	Normal distribution $\mathcal{N}(1.5, 0.3)$
P, Q	Level of background excitation in the excitatory, inhibitory subpopulation	0.9, 0	0, 0
c_1, c_3	Average number of excitatory synapses per cell	16, 15	$\mathcal{N}(16, 1.5)$, 15
c_2, c_4	Average number of inhibitory synapses per cell	12, 3	$\mathcal{N}(12, 1.0)$, 3
$\theta_e, \theta_i, a_e, a_i$	Maximal slope parameters of the logistic curve for the excitatory, inhibitory subpopulation	4, 3.7, 1.3, 2	4.5, 3.7, 1.3, 2
μ_1, μ_2	Modulation time constants	0.002, 0.002 (ms^{-1})	0.001, 0.001 (ms^{-1})
κ_ϕ	Sensitivity to the variations of the metabolic substrate	75 (substrate^{-1})	75 (substrate^{-1})
κ_β	Sensitivity to the variations of the neuronal activity	4.5 (rate^{-1})	4.5 (rate^{-1})
η, ζ	Threshold values to changes in the metabolic substrate, slow gating process	25, 0.27 (substrate)	25, 0.27 (substrate)
k_r, k_c	Metabolic recovery and consumption rates	0.90, 0.70 ($\text{substrate}/\text{ms}$)	$\mathcal{N}(0.50, 0.10)$, $\mathcal{N}(0.90, 0.10)$
v_1, v_2	Homeostatic autoregulation time constants	0.08, 0.08 (s^{-1})	0.08, 0.08 (s^{-1})

k_c) leads to the manifestation of sustained cycles in the slow subsystem, resulting in bursts of fast neuronal activity (i.e., burst suppression). The dynamical mechanism is illustrated in the inset of Fig. 2, noting the fall of $M(t)$ and $\phi(t)$ through the course of each burst. During suppression epochs, the substrate process $M(t)$ recovers until $\phi(t)$ can again facilitate the regeneration of a new oscillatory burst.

3.2 Decrease in neuronal activity leads to burst suppression-like activity

We now study a new feature of the model—the ability to produce bursting due only to downregulation of neural activity, without any other extrinsic alteration to the slow subsystem. Specifically, we focus on the parameter c_2 , the strength of inhibitory to excitatory coupling in the Wilson–Cowan equations. Potentiation of this parameter

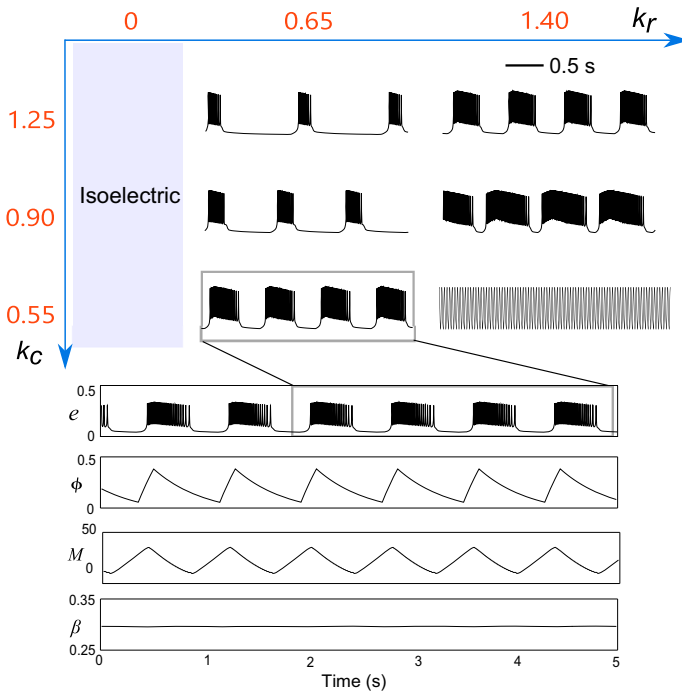


Fig. 2 Direct manipulation of the substrate recovery and consumption rates leads to anticipated burst-like dynamics. The *inset* shows detailed time evolution of the model variables for the case when the model produces cycles in the slow subsystem, i.e., burst suppression

serves as an analogue for the actions of inhibitory agonists, such as anesthetic drugs that potentiate the γ -aminobutyric acid (GABA) synaptic current (Brown et al. 2010; Ching and Brown 2014). We assume that the self-inhibitory parameter c_4 is not sensitive to these GABAergic effects, owing to the relatively smaller proportion of inhibitory neurons within cortical columns (McCarthy et al. 2008).

Figure 3a shows an example of the model response as c_2 is increased in a ramp-like fashion as a function of time. We observe that the system manifests burst-like slow cycles that gate the faster neural activity. As c_2 first increases, an initial transient suppression occurs, following which the amplitude of bursts progressively decreases and the duration of suppression becomes progressively longer. Conceptually, these dynamics can be interpreted as follows: as excitatory neural activity is sufficiently inhibited (i.e. attenuated amplitude of e), the rate variable β (which, in effect, scales k_r) homeostatically decreases, thus leading to the same basic cycling mechanism discussed above.

The homeostatic coupling is critical to the manifestation of bursting as a function of increasing c_2 . Indeed, if the homeostatic mechanism is disabled and β is fixed, the fast subsystem amounts to a two-dimensional Wilson–Cowan model extrinsically forced by the slow subsystem. That is, the slow subsystem provides an essentially fixed supply of metabolic substrate to the fast system. In this scenario, it is impossible

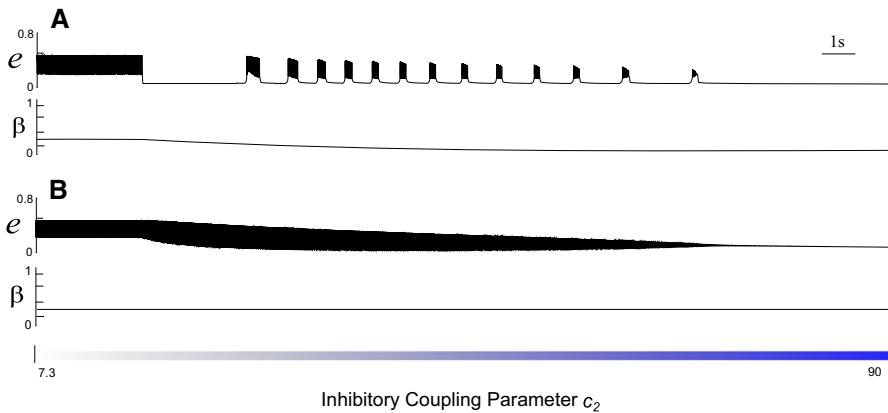


Fig. 3 Model response when inhibitory coupling parameter c_2 is increased as a function of time. **a** The homeostatic mechanisms enable neural modulation via c_2 to create burst-like activity, without any other extrinsic alteration to the slow subsystem. **b** When the homeostatic coupling is disabled (i.e., with $\beta = 0.3$ fixed), the amplitude of neuronal activity gradually decreases without the appearance of bursts

for a downregulation of fast activity to alter the dynamics of this fixed supply and bursting would only occur if the slow subsystem was already in an oscillatory regime, i.e., Sect. 3.1 (Fig. 2) above. If the slow subsystem were not already in an oscillatory regime, an increase in c_2 would be expected to simply cause a gradual attenuation in excitatory activity amplitude. This is demonstrated in Fig. 3b, wherein the homeostatic coupling is disabled and β is fixed at a constant value ($\beta = 0.3$). As c_2 increases the amplitude of the excitatory activity attenuates gradually, but no bursting arises. Thus, the homeostatic coupling is necessary for the emergence of burst-like dynamics, which we explore in detail via bifurcation analysis, below.

Bifurcation analysis To study these dynamics, we performed a bifurcation analysis on the model with respect to c_2 , summarized through the diagram in Fig. 4 (see Appendix A for details). This diagram depicts the dynamical behavior of the slow variable ϕ as a function of c_2 . Several observations can be made from this analysis.

First, for low values of c_2 , the variable ϕ exhibits a constant steady state ($\phi(t) \approx 0$). In this regime, the fast subsystem is able to sustain continuous oscillatory activity, such that the entire system exhibits a stable limit cycle corresponding to oscillations in the fast subsystem.

At the critical point around $c_2 = 7.7$, the system exhibits a homoclinic bifurcation, wherein an unstable and saddle node emerge, enveloped by a stable limit cycle. In the parameter regime $c_2 \in [7.7, 8.6]$ the system is bistable (stable node/stable limit cycle). Another bifurcation occurs at $c_2 = 8.6$, wherein the stable and saddle nodes are abolished, leaving the limit cycle as the only stable solution of the system. The cycling in ϕ remains until $c_2 = 69$, at which point ϕ again manifests a stable steady state via a Hopf bifurcation. For values of $c_2 > 69$, the neuronal activity becomes quiescent, such that the entire system exists at a stable fixed point.

Figure 5 illustrates the system trajectories for three different values of c_2 , corresponding to different points along the bifurcation diagram. Note especially that for

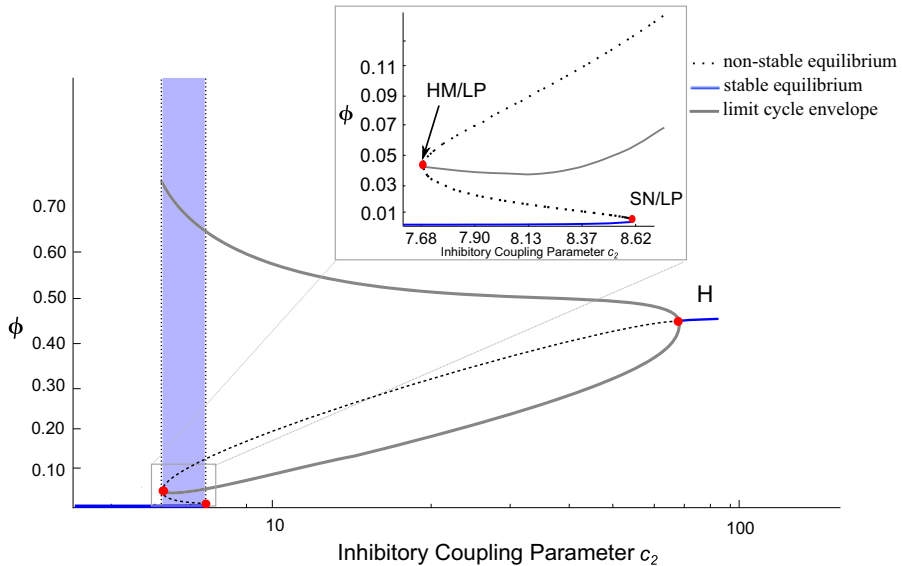


Fig. 4 Bifurcation diagram of the model with respect to the inhibitory coupling parameter c_2 . (HM/LP) Homoclinic/limit point bifurcation. (SN) Saddle node bifurcation. (H) Hopf bifurcation. The blue lines represent stable equilibrium points. Non-stable equilibrium points (unstable equilibrium points or saddle nodes) are marked by black dashed lines. The gray curves are the envelope of the periodic limit cycle. The blue shaded region indicates values of c_2 for which the model is bistable (note detailed inset) (color figure online)

a low value of c_2 , the fast system produces a sustained oscillation that resides on a fixed ϕ plane (as indicated by the bifurcation diagram). For a high value of c_2 (to the right of the Hopf bifurcation), ϕ is again fixed, however, in this case, fast cycles are abolished.

The bifurcation analysis provides a concise dynamical interpretation of the transition from continuous neuronal activity into burst suppression with respect to increasing neuronal inhibition. It demonstrates the emergence of burst suppression-like dynamics with respect to a change only in the neuronal subsystem, without requiring an extrinsic modulation of the substrate recovery or consumption rates. In addition to this basic characterization, a number of more nuanced dynamical features can be inferred.

Bistability and hysteresis Characteristically of fast-slow systems of this type, the model predicts an interesting dynamical feature not previously studied in the context of burst suppression, namely, the presence of bistability in the blue shaded region of Fig. 4. In this region, the model gating variable ϕ may exhibit at stable steady state (leading to continuous fast activity), or stable cycles (leading to burst suppression). Consequently, the manifestation of burst suppression exhibits hysteresis with respect to c_2 , wherein the thresholds for induction and emergence from burst suppression are different. Hysteresis and bistability have been described in the context of general anesthesia generally (Friedman et al. 2010; Joiner et al. 2013), but have not been studied for burst suppression specifically. The presence of this phenomenon is, thus, a potentially verifiable model prediction.

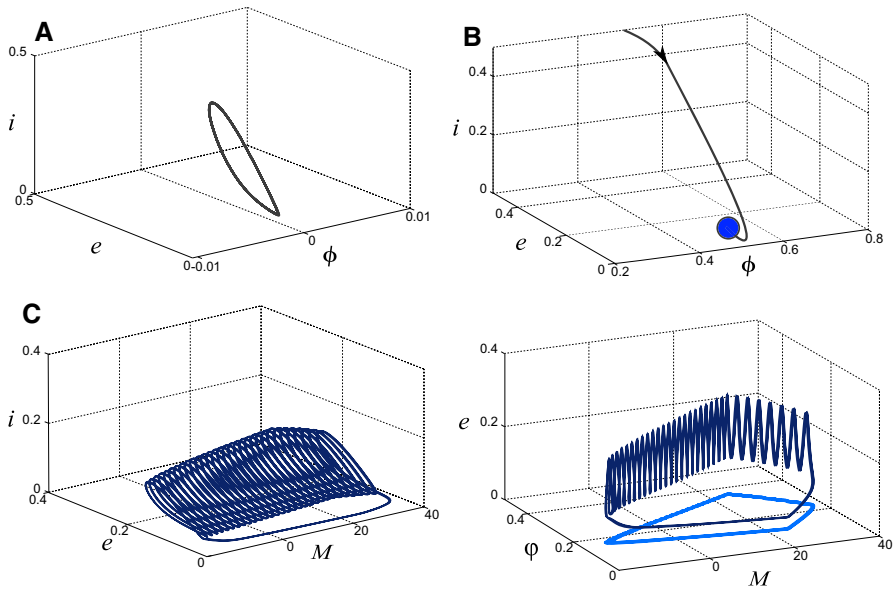


Fig. 5 Fast-slow interactions in different dynamical regimes with respect to the inhibitory coupling parameter c_2 . **a** $c_2 = 7.3$: the system exhibits a periodic solution, where the fast subsystem oscillates within the plane $\phi = 0$. **b** $c_2 = 80$: the entire system resides at a stable fixed point, with $\phi \approx 0.47$. **c** $c_2 = 14$: the system exhibits burst suppression, with suppression characterized by regeneration of the substrate M (left). The projection of the (e, ϕ, M) locus on the (ϕ, M) plane (the blue trace, right) corresponds to the limit cycle portrayed in Fig. 4 (color figure online)

Model predicts relative changes in burst suppression lengths The model provides agreement with a number of more subtle features of burst suppression, including a progressive reduction in the overall amplitude of bursts as a function of neuronal inactivation (Fig. 6a). It also predicts that the average fraction of time spent in suppression, the so-called ‘burst suppression ratio’ (BSR = 0, continuous activity; BSR = 1, complete isoelectricity), has an effective lower bound. The implication is that as c_2 increases, suppression will manifest with a minimal length; i.e., arbitrarily short suppression lengths cannot occur (Fig. 6b). It is important to note that the homoclinic bifurcation (in Fig. 4, at $c_2 = 7.7$) implies that the slow cycles themselves can approach infinite periodicity. However, since ϕ is additive to P in (3), the gating mechanism necessitates a certain minimal amplitude of ϕ , below which fast activity is not facilitated. This, in combination with the extrinsic noise, ensures the lower bound effect observed in Fig. 6b.

Further, as c_2 increases, the BSR increases asymptotically to 1, with suppression getting progressively longer and bursts becoming shortened and weaker (e.g., Figs. 3, 6b–d). In this figure, the blue shaded regions indicate values of c_2 for which the model is bistable.

Note that these temporal manifestations are emergent properties of the model insofar as we did not tailor any of the model functions to specifically produce such phenomena. The shape of the curves in Fig. 6a–d are consistent with experimental characterizations of burst suppression that report similar changes in burst and suppression durations and amplitudes as functions of neuronal circuit downregulation (see Sect. 5 below).

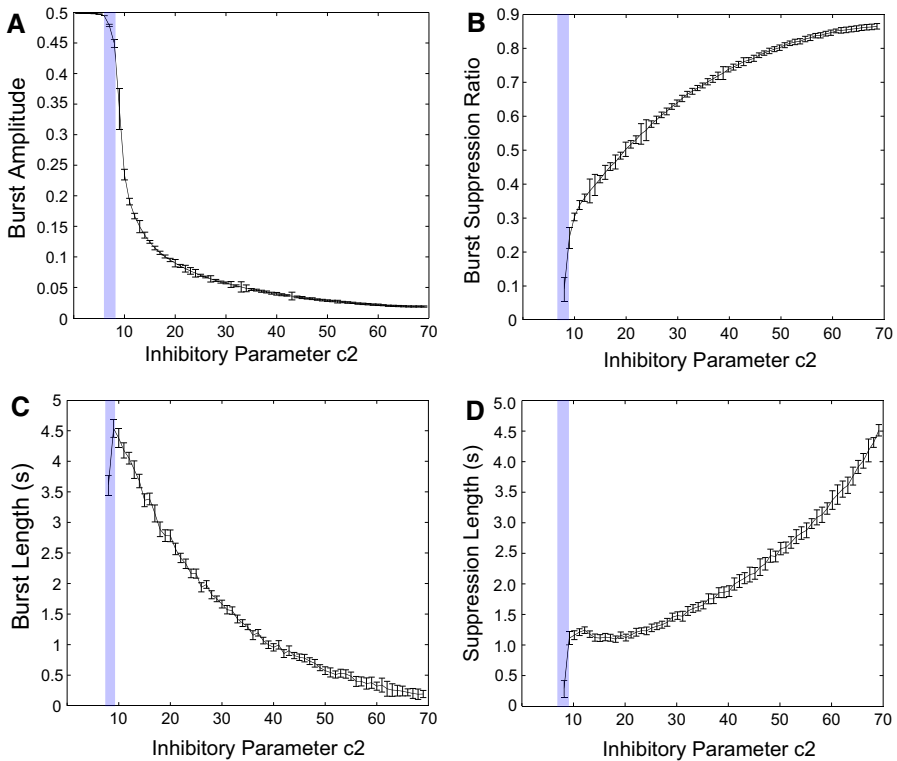


Fig. 6 **a** Burst amplitude, **b** burst suppression ratio (BSR), **c** burst duration and **d** suppression duration, are plotted as functions of c_2 . A small amount of extrinsic noise is included via $W_j^e(t)$, $W_j^i(t)$ in (4). The blue shaded region indicate values of c_2 for which the model is bistable. Mean and standard error over $n = 100$ realizations are plotted (color figure online)

Homeostatic relationship between ϕ and fast dynamics Finally, it is interesting to note that, for larger $c_2 > 69$, $\phi(t)$ actually assumes a larger value at equilibrium than when $c_2 < 7.7$. Thus, despite the fast subsystem being quiescent, the gating variable is actually more facilitating than when the fast system was able to sustain continuous activity. Recall, however, that the bifurcation variable here is c_2 , the intra-column inhibitory coupling strength. Thus, for low values of c_2 the column itself is less inhibited and consequently, is relatively insensitive to the gating variable ϕ . Indeed, the presence of homeostatic dynamics leads to a non-monotonic relationship between the magnitude of ϕ (in absolute terms) and c_2 .

4 Synchronization analysis and network phenomena

We now proceed with a more rigorous analysis of the dynamics of the model when embedded in interconnected networks. Specifically, we explore how the model might reconcile classical descriptions of burst suppression as a highly synchronous, widespread, whole-brain phenomenon (Clark and Rosner 1973), versus more recent characterization that reveals a substantial amount of spatial heterogeneity in timing of

burst onset and offset. Our investigation centers on the roles of inter-column coupling and dynamical heterogeneity in mediating burst synchronization.

4.1 Burst synchronization due to neural coupling

We first study two reciprocally coupled columns, with coupling restricted to the fast subsystem. Thus, Eqs. (3) and (4) reduce to

$$\begin{aligned}\dot{e}_j = & -e_j + (k_e - r_e e_j) \mathcal{F}[c_1 e_j - c_2 i_j + k_j^{fe} e_k + P + \phi_j(t)] \\ & + W_j^e(t)\end{aligned}\quad (10)$$

$$\dot{i}_j = -i_j + (k_i - r_i e_j) \mathcal{F}[c_3 e_j - c_4 i_j + k_j^{fi} e_k + Q] + W_j^i(t), \quad (11)$$

where k_j^{fe} is the coupling strength. Note that, with modification to c_1 , diffusive coupling between excitatory populations can be formulated in much the same way: $c_1 e_j + k_j^{fe}(e_k - e_j) = (c_1 - k_j^{fe})e_j + k_j^{fe}e_k$. We are interested in the extent to which coupling of this form might cause bursts to synchronize, i.e., without necessarily inducing synchronization of the fast activity. To be concrete, we define this notion as follows:

Definition 1 (*Burst synchronization*) For columns j, k , burst synchronization occurs if $\forall \epsilon > 0, \exists \mathfrak{T}$ such that $\forall t > \mathfrak{T}, \|\phi_k(t) - \phi_j(t)\| < \epsilon$. Here, $\|x(t)\| := \sup_t |x(t)|$.

Proposition 1 Consider two columns coupled in the form (10). The substrate rates $k_r > 0$ and $k_c > 0$ may be different for each column, but the two columns are otherwise homogeneously parametrized. There exists a coupling strength $k_f^* > 0$ so that for $k_j^{fe}, k_k^{fe} > k_f^*$, burst synchronization occurs.

Proof See Appendix B.

Perhaps intuitively, this result indicates that two columns with homogeneously parametrized fast subsystems (but potentially different homeostatic coupling rates), each producing burst suppression-like activity, will produce synchronized bursts provided that coupling is sufficiently large.

Manifestation of synchronization in larger networks The analysis predicts that synchronous bursting will manifest in local cortical regions, wherein nearby columns are similar, and coupling strength is relatively high. In contrast, over longer distances, heterogeneity and attenuation of excitation would lead to variation in the onset and offset of bursting. We observe such an effect in simulations of larger networks. For example, in Fig. 7 we simulated 36 columns, where each column was coupled to its nearest neighbors in a spherical grid lattice. For each column, k_c was obtained as a realization of a normal random variable with mean 0.90 and standard deviation 0.10. Similarly, k_r was chosen from $\mathcal{N}(0.50, 0.10)$. In general the values of k_j^{fe}, k_j^{fi} would be expected to be different, since the excitatory population would typically receive long range cortico-cortical synaptic connections. Here, we make a simplifying approximation and draw them independently, but from the same distribution, $\mathcal{N}(\bar{k}_f, 0.3)$. The

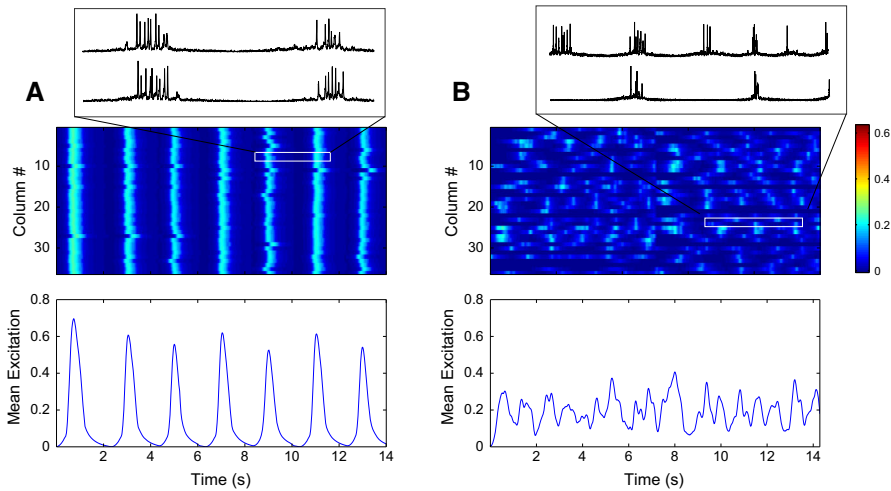


Fig. 7 Synchronization of bursting activity in a model of coupled columns with heterogeneous parameters and neuronal coupling of the form (10). *Top* excitation (*color*) as a function of time for each column. *Bottom* average excitation. **a** Strong coupling ($\bar{k}_f = 2.5$). **b** Weak coupling ($\bar{k}_f = 1.5$). Note that the fast (within-burst) activity is not synchronized across columns in either case (color figure online)

initial conditions for e, i are taken as realizations of a uniform random variable over the range $[0, 0.5]$, ϕ, β are initially chosen from a uniform distribution over $[0, 1]$ and M from a uniform distribution over $[0, 30]$. The variances were chosen to be around 25–30 % of the mean in all cases, based on previous discussions of neural variability (Marder and Goaillard 2006). All parameters are chosen as independent realizations, which is a simplifying approximation since, in reality, some spatial correlation would be expected. Further, for each column we specified an extrinsic white noise processes $W_j^e(t), W_j^i(t)$ of zero mean and variance 0.04.

For strong coupling, $\bar{k}_f = 2.5$, the columns tend to exhibit relatively widespread bursting consistent with the result in Proposition 1 (Fig. 7a). In contrast, for weaker coupling $\bar{k}_f = 1.5$, bursting is substantially more local and the network as a whole is less synchronized (Fig. 7b).

4.2 Burst synchronization due to substrate interactions

We further consider coupling mediated through diffusive interactions in the substrate dynamics. Here, for two columns $j, k, j \neq k$, Eqs. (7)–(8) become

$$\dot{M}_j = k_r \beta_j - k_c \left(\frac{e_j^2}{0.01 + e_j^2} \right) + k_j^s (M_k - M_j) \quad (12)$$

where k_j^s is a diffusive constant. It is perhaps intuitive that such coupling would promote burst synchronization, a premise that is readily shown:

Proposition 2 Consider two columns coupled in the form (12). The substrate rates $k_r > 0$ and $k_c > 0$ may be different for each column, but the two columns are otherwise homogeneously parametrized. There exists a coupling strength $k_s^* > 0$ so that burst synchronization occurs for $k_j^s, k_k^s > k_s^*$.

Proof See Appendix C.

Thus, again, substantial burst synchronization is expected locally, within regions in which substrate mixing is significant. Figure 8 depicts a simulated output of 36 columns, again composed in a nearest neighbor topology with the same heterogeneity in columnar parameterization and initial conditions as described above for Fig. 7. Here, no neuronal coupling is included in the network and we study only substrate interactions. We contrast the nature of burst synchronization for strong (Fig. 8a, k_j^s drawn from $\mathcal{N}(0.10, 0.015)$) and weak (Fig. 8b, k_j^s drawn from $\mathcal{N}(0.01, 0.005)$) interactions. The outcome of Proposition 2 is clearly seen, with substantially more global burst synchronization associated with stronger interactions.

4.3 Heterogeneity, noise and local versus global bursting

The propositions provide two complementary pathways to burst synchronization: via neuronal interactions and via mixing of substrate between columns. When both are present simultaneously, the overall network disposition would be expected to be further biased towards synchronization. Thus, we can infer that burst synchronization is

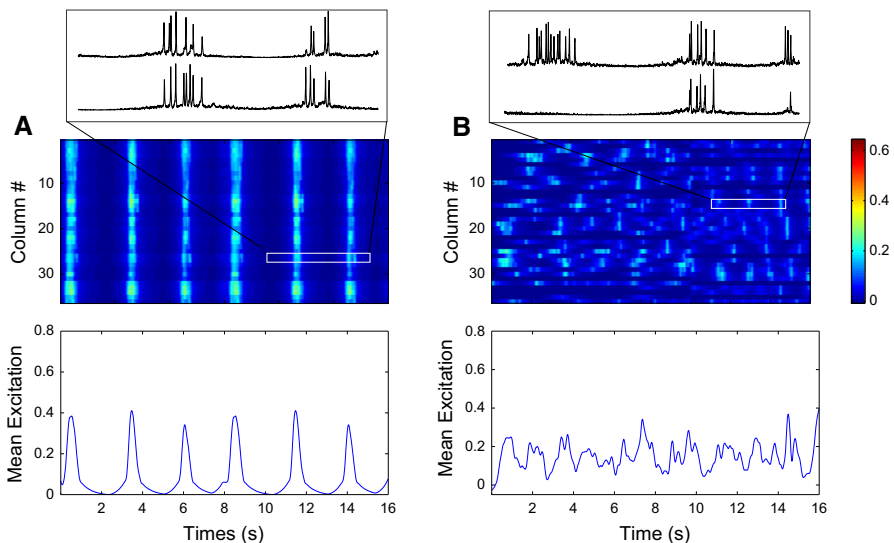


Fig. 8 Synchronization of bursting activity in a model of coupled columns with heterogeneous parameters and diffusive coupling in the substrate dynamics of the form (12). *Top* excitation (color) as a function of time for each column. *Bottom* average excitation. **a** Strong coupling ($k_s = 0.10$). **b** Weak coupling ($k_s = 0.01$). Note that the fast (within-burst) activity is not synchronized across columns in either case (color figure online)

pervasive, however, due to intrinsic heterogeneity, is more prevalent in local regions. Indeed, truly ‘global’ bursts would require strong coupling to be expressed broadly throughout the network, which is not consistent with physiology. Nevertheless, it could be the case that strong surges of afferent excitation might promote brief epochs of widespread activity.

Thus, to complete our simulation study, we consider an additional example of a 36-column network with the same neuronal coupling type as in Fig. 7 with two new sources of potential heterogeneity. Specifically, whereas in our previous analysis and examples we considered spatial heterogeneity in the slow subsystem, we now consider it exclusively in the fast dynamics. Further, we consider a random connectivity in the form of a small-world network topology. To achieve these sources of heterogeneity, the excitatory–excitatory coupling c_1 was chosen from $\mathcal{N}(16, 1.5)$ and the inhibitory–excitatory coupling c_2 from $\mathcal{N}(12, 1.0)$. In addition, starting from the nearest neighbor configuration, connections were randomly rewired with a probability of 0.15. Again, to each column we add an independent white noise process (i.e., $W_j^e(t)$, $W_j^i(t)$ in (4)) with zero mean and variance 0.04. It is readily evident that bursts still tend to locally synchronize, agreeing with the observation of Fig. 7. However, widespread, globally synchronized bursts do appear sporadically when the connection strength is moderately increased (i.e., Fig. 9b, wherein both local and global bursting is observed) and particularly when connections are strong (i.e., Fig. 9c).

Thus, the model predicts that burst synchronization is primarily expressed in local, more strongly connected sub-regions; while global bursts are rarer events mediated by simultaneous elevation in the excitability of columns across the network.

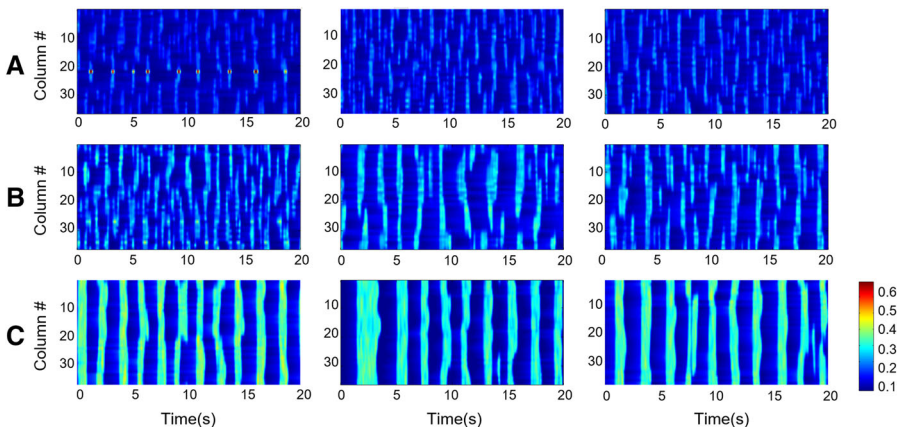


Fig. 9 Local versus global bursting in heterogeneous networks with small-world topology. *Each row depicts three different random network realizations: a* burst synchronization is mostly local for weak coupling ($\bar{k}_f = 1.5$). *b* Both local and global burst synchronies are manifest when coupling is increased ($\bar{k}_f = 2.0$). *c* Global burst synchronization is dominant when coupling is strong ($\bar{k}_f = 3.5$)

5 Discussion and conclusions

In this paper, we have introduced a low-dimensional model for burst suppression based on the premise of homeostatic dynamics of neuronal substrate interaction. In the model, which takes the form of a fast-slow dynamical system, burst suppression-like activity can be induced through extrinsic regulation of substrate recovery or, alternatively, through downregulation of neuronal activity without any direct effect on the substrate dynamics. The latter feature is consistent with the primary actions of anesthetic drugs, and thus closes a conceptual gap in earlier computational models of burst suppression (Ching et al. 2012; Liley and Walsh 2013), wherein perturbation was exclusive to the slow (substrate) system. The modeled substrate variables ϕ , M , β , while deliberately abstract, can be readily placed into the context of, for example, the ATP-Sodium exchange dynamics developed in Ching et al. (2012).

Homeostatic interactions between fast and slow dynamics in burst suppression have also been explored in Bojak et al. (2015). There, the key homeostatic variable is the maximal amplitude of post-synaptic potentials, which was a fixed parameter in Liley and Walsh (2013). Anesthesia is modeled in Bojak et al. (2015) as an attenuation in these post-synaptic potentials. However, this variable appears in both fast and slow subsystems. Thus, anesthetic-induced neural modulation enters as a multiplicative factor in both the fast *and* slow subsystems. In contrast, here, we have placed emphasis on neuronal-substrate interactions wherein modulation of *only* the fast dynamics can lead to burst suppression.

Dynamically, our model exhibits an ‘parabolic’ bursting mechanism (Ermentrout and Terman 2010), similar to generic fast-slow burst mechanisms in cell-level neuronal models (Rinzel 1987; Izhikevich 2007). The relative low-dimensionality and deliberate generality of our model, coupled with its observed robustness, thus suggests that burst suppression is due to a basic set of dynamical principles, and not on our specific functional forms.

In addition to offering an explanation for a pervasive mechanistic pathway to burst suppression, the model is consistent with new characterizations of burst and suppression morphology (Kenny et al. 2014; Westover et al. 2015). It also exhibits emergent bistability that lends potential dynamical insight into putative hysteresis in neuronal responses to inhibition (e.g., via anesthetic drugs Friedman et al. 2010; Joiner et al. 2013). Further, the model enables a dynamical analysis of burst synchronization, towards reconciling sources of asynchrony in burst onset and offset (Henry and Scoville 1952; Lazar et al. 1999). It is shown that both neuronal coupling and substrate interactions can lead to burst synchronization, but that intrinsic heterogeneity leads primarily to local expression of synchronization. Nevertheless, afferent excitation may lead to occasional widespread (global) bursting events, consistent with the recent finding of Lewis et al. (2013). Such an interpretation may be complementary to other models for localized burst suppression in the cortex, whose mechanisms center on spreading excitation and wave propagation (Bojak et al. 2015).

Appendix A: A computational details

All simulations and subsequent computations were performed in the MATLAB (Natick, MA) numerical environment. A standard four step Runge-Kutta algorithm was used for evaluating differential equations. Bifurcation analysis was completed using the MATCONT package (Dhooge et al. 2003). Since the model is dimensionless, a time-base was specified in order to be consistent with experimental characterizations of burst suppression.

Appendix B: Proof of Proposition 1: synchronization due to neural coupling

In an isolated column, define the solution of the substrate equation as

$$M(t) = k_r \mathcal{B}(t) - k_c \mathcal{C}(e, \min(t, \tau)) \quad (13)$$

where τ is the burst duration in a burst-suppression cycle with period T . Note that the consumption term $\mathcal{C}(e, t)$ increases monotonically through the course of a burst and, thus, assumes its maximum at the time of burst termination, τ . Thus,

$$\begin{aligned} d\mathcal{C}(e, t)/dt &= \frac{e(t)^2}{0.01 + e(t)^2}, \\ d\mathcal{B}(t)/dt &= \beta(t), \end{aligned} \quad (14)$$

respectively. Observe that $\mathcal{C}(e, t)$ is monotonically increasing with respect to $e(t)$ and t .

Suppose that the columns j, k are coupled via the fast neuronal coupling as in (10). Assume without loss of generality that $\kappa = 1$, then a necessary and sufficient condition for burst synchronization is that

$$\dot{\phi}_\perp = \mu_1(-\phi_\perp) + \mu_2 \left(\frac{\exp M_j - \exp M_k}{(1 + \exp M_j)(1 + \exp M_k)} \right) \quad (15)$$

has a steady state at $\phi_\perp(t) = 0$, where $\phi_\perp(t) = \phi_j(t) - \phi_k(t)$. Note here that for notational convenience, we have dropped the time dependence on M_j, M_k .

Notice Eq. (15) has a stable steady state solution at $\phi_\perp = 0$ if and only if $\|\exp(M_j) - \exp(M_k)\| \rightarrow 0$. By monotonicity of the exponential function, this condition can be reduced to $\|M_j\| = \|M_k\|$. From Definition 1 and (13)–(14),

$$\left| k_r^j \mathcal{B}_j(T) - k_c^j \mathcal{C}(e_j, \tau) \right| = \left| k_r^k \mathcal{B}_k(T) - k_c^k \mathcal{C}(e_k, \tau) \right| \quad (16)$$

Since the substrate variables are nonnegative, rearranging the terms above and normalizing by k_c^k yields

$$\mathcal{C}(e_k, \tau) - k'_c \mathcal{C}(e_j, \tau) = \frac{k_r^k \mathcal{B}_k(T) - k_r^j \mathcal{B}_j(T)}{k_c^k}, \quad (17)$$

where $k'_c = k_c^j / k_c^k > 0$. Note that $\mathcal{B}_{j,k}(T)$ remain dependent on the coupling strength $k_{j,k}^f$.

For notational convenience, define $g(e_k) \equiv \mathcal{C}(e_k, \tau)$, $g(e_j) \equiv \mathcal{C}(e_j, \tau)$ and let $B_{k,j}(k_j^f, k_k^f)$ denote the right hand side of (17), i.e.,

$$B_{k,j}(k_j^f, k_k^f) = \frac{k_r^k \mathcal{B}_k(T) - k_r^j \mathcal{B}_j(T)}{k_c^k}. \quad (18)$$

Lemma 1 Suppose that $k_j^f = k_k^f \equiv k_f$. Define $\rho_j(t, k_f) = \phi_j(t) + k_f e_k(t) - P_j$. There exists an interval $[k_1, k_2]$, $0 \leq k_1 < k_2$, so that $B_{k,j}(k_f) = g(\rho_k(k_f)) - k'_c g(\rho_j(k_f))$ for $k_f \in [k_1, k_2]$, and then $k_f^* \equiv k_1$ is the threshold value for synchronization via fast neuronal coupling.

Proof It is a direct consequence of Theorem 3 in Wilson and Cowan (1972) that there exists an invertible mapping between e_j and ρ_j on a time interval around $nT_j + \tau_j$, where T_j and τ_j are the period and duration, respectively, of bursting in column j . Thus, an equivalent condition to Eq. (17) can be formulated as:

$$g(\rho_k(k_f)) - k'_c g(\rho_j(k_f)) = B_{k,j}(k_f) \quad (19)$$

where $k_f \geq 0$. We seek a k_f^* so that $k_f > k_f^*$ is a solution to the above equation.

Note that $\rho_j(k_f) = \phi_j + k_f e_k - P_j$ is an increasing function of the neuronal coupling coefficient k_f at time $nT_j + \tau_j$. By the Jordan decomposition Theorem (Royden 1969), for some finite interval $k_f \in [k_1, k_2]$, $B_{k,j}(k_f)$ can be expressed as difference of nondecreasing functions: $B_{k,j}(k_f) = v_1(k_f) - v_2(k_f)$. Further, the total variation $TV(B_{k,j}(\cdot)) = v_1(k_2) - v_1(k_1) + v_2(k_2) - v_2(k_1)$. The pair v_1, v_2 is unique up to addition of a constant.

Since $0 \leq \phi \leq 1$, $0 \leq e \leq 1/2$, there exist integers $0 < N_1 < N_2 < \infty$ so that $\rho_k(k_2) = k_2/2 - P - 2^{-N_1}$, $\rho_j(k_2) = k_2/2 - P - 2^{-N_2}$ for large k_2 . After some algebraic manipulation, we thus obtain

$$\begin{aligned} k'_c g(\rho_j(k_1)) + g(\rho_k(k_1)) &= g(k_2/2 - 2^{-N_1}) \\ &+ k'_c g(k_2/2 - 2^{-N_2}) - TV(B_{k,j}(\cdot)) \end{aligned} \quad (20)$$

and

$$\begin{aligned} k'_c g(\rho_j(k_1)) - k'_c g(\rho_k(k_1)) &= B_{k,j}(k_2) - B_{k,j}(k_1) \\ &+ g(k_2/2 - 2^{-N_2}) - g(k_2/2 - 2^{-N_1}) \end{aligned} \quad (21)$$

Equations (20) and (21) are linear equations with respect to $g(\rho_k(k_1))$, $g(\rho_j(k_1))$. Since $k'_c > 0$, $TV(B_{k,j}(\cdot))$ and $B_{k,j}(k_2) - B_{k,j}(k_1)$ are finite, a solution for $g(\rho_k(k_1))$, $g(\rho_j(k_1))$ exists. Thus, since $g(\cdot)$ is positive and monotonically increasing, k_1 exists and $k_f^* = k_1$ is the threshold value for burst synchronization.

The proposition follows as a direct extension of Lemma 1, noting that if $k_j^f \neq k_k^f$, (19) becomes $g(\rho_k(k_k^f)) - k'_c g(\rho_j(k_j^f)) = B_{k,j}(k_k^f, k_j^f)$ and, consequently, $k_k^f, k_j^f \in [k_f^*, k_2]$ is sufficient to ensure burst synchronization.

Appendix C: Proof of Proposition 2: synchronization due to substrate coupling

In an analogous manner to the proof of Proposition 1, we will show that $\phi_\perp = \phi_j(t) - \phi_k(t)$ tends to zero as $t \rightarrow \infty$.

Lemma 2 Assume that $k_j^s = k_k^s = k_s$. There exists a k_s^* , so that for $k_s > k_s^*$, burst synchronization occurs.

Proof We follow the analysis paradigm outlined in Dhamala et al. (2004), exploiting the separation of time scales between the fast and slow subsystems. Define the transverse equations for the slow dynamics as:

$$\begin{aligned}\dot{\phi}_\perp &= \dot{\phi}_j(t) - \dot{\phi}_k(t) \\ &= \mu_1(-\phi_\perp) + \mu_2 \left(\frac{\exp M_j - \exp M_k}{(1 + \exp M_j)(1 + \exp M_k)} \right); \\ \dot{M}_\perp &= \dot{M}_j(t) - \dot{M}_k(t) \\ &= -2k_s M_\perp + k_r^j \beta_j - k_r^k \beta_k + k_c^k y(e_k) - k_c^j y(e_j); \\ \dot{\beta}_\perp &= \dot{\beta}_j(t) - \dot{\beta}_k(t) \\ &= \nu_1(-\beta_\perp) + \nu_2 \left(\frac{\exp e_j - \exp e_k}{(1 + \exp e_j)(1 + \exp e_k)} \right); \quad (22)\end{aligned}$$

where $y(e_{k,j}) = \frac{e_{k,j}^2}{0.01 + e_{k,j}^2}$. Linearizing (22) yields the following sufficient condition for asymptotic attractiveness of the subspace $\{(e_\perp, i_\perp, \phi_\perp, M_\perp, \beta_\perp) \in \mathbb{R}^5 : (\phi_\perp, M_\perp, \beta_\perp) = (0, 0, 0)\}$:

$$-2k_s + k_r^j \frac{d\beta_j}{dM_\perp} - k_r^k \frac{d\beta_k}{dM_\perp} + k_c^k \frac{dy(e_k)}{dM_\perp} - k_c^j \frac{dy(e_j)}{dM_\perp} < 0. \quad (23)$$

Using the same notation as in (13), we obtain

$$\begin{aligned}M_\perp(t) &= -2k_s \int_t M_\perp(t) dt + k_r^j \mathcal{B}_j(t) \\ &\quad - k_c^j \mathcal{C}(e_j, \min(t, \tau_j)) - k_r^k \mathcal{B}_k(t) + k_c^k \mathcal{C}(e_k, \min(t, \tau_k)) \quad (24)\end{aligned}$$

and, further,

$$\begin{aligned} \left. \frac{dM_{\perp}(t)}{d\beta_j(t)} \right|_{M_{\perp}(t)=0} &= \left. \frac{-2k_s d \int_t M_{\perp}(t) dt}{d\beta_j(t)} \right|_{M_{\perp}(t)=0} \\ &\quad + \left. \frac{dM_{\perp}(t) dt}{d^2 \mathcal{B}_j(t)} \right|_{M_{\perp}(t)=0} = k_r^j / \beta_j(t), \end{aligned} \quad (25)$$

$$\begin{aligned} \left. \frac{dM_{\perp}(t)}{d\beta_k(t)} \right|_{M_{\perp}(t)=0} &= \left. \frac{-2k_s d \int_t M_{\perp}(t) dt}{d\beta_k(t)} \right|_{M_{\perp}(t)=0} \\ &\quad + \left. \frac{dM_{\perp}(t) dt}{d^2 \mathcal{B}_k(t)} \right|_{M_{\perp}(t)=0} = k_r^k / \beta_k(t), \end{aligned} \quad (26)$$

Note, also, that

$$\begin{aligned} \left. \frac{dM_{\perp}(t)}{dy(e_j)} \right|_{M_{\perp}(t)=0} &= \left. \frac{-2k_s d \int_t M_{\perp}(t) dt}{dy(e_j)} \right|_{M_{\perp}(t)=0} \\ &\quad + \left. \frac{dM_{\perp}(t) dt}{d^2 \mathcal{C}(e_j, \min(t, \tau_j))} \right|_{M_{\perp}(t)=0} = k_c^j / y(e_j), \end{aligned} \quad (27)$$

$$\begin{aligned} \left. \frac{dM_{\perp}(t)}{dy(e_k)} \right|_{M_{\perp}(t)=0} &= \left. \frac{-2k_s d \int_t M_{\perp}(t) dt}{dy(e_k)} \right|_{M_{\perp}(t)=0} \\ &\quad + \left. \frac{dM_{\perp}(t) dt}{d^2 \mathcal{C}(e_k, \min(t, \tau_k))} \right|_{M_{\perp}(t)=0} = k_c^k / y(e_k), \end{aligned} \quad (28)$$

Substituting Eqs. (25)–(28) in Eq. (23) yields:

$$k_s > \frac{1}{2} \sup_t |\beta_j(t) - \beta_k(t) - y(e_j(t)) + y(e_k(t))| \quad (29)$$

Notice, however, that $\beta_{j,k}$ and $y(e_{j,k})$ are nonnegative:

$$\begin{aligned} |\beta_j(t) - \beta_k(t) - y(e_j) + y(e_k)| &\leq |\beta_j(t) + y(e_k)| \\ &\leq |\beta_j(t)| + |y(e_k)|. \end{aligned} \quad (30)$$

Also $\beta_{j,k}$ and $y(e_k)$ are bounded by 1 (because $e_{j,k}$ are bounded), so:

$$\begin{aligned} \frac{1}{2} \sup |\beta_j(t) - \beta_k(t) - y(e_j) + y(e_k)| &\leq \frac{1}{2} (|\beta_j(t)| + |y(e_k)|) \\ &< \infty \end{aligned} \quad (31)$$

The lemma follows directly from (29) and (31), noting that, for instance, $k_s^* = 1$ is sufficient to ensure the result.

The proposition then follows from Lemma 2, noting that if $k_j^s \neq k_k^s$, $\dot{M}_\perp = -(k_j^s + k_k^s)M_\perp + k_r^j \beta_j - k_r^k \beta_k + k_c^k y(e_k) - k_c^j y(e_j)$ and any values of k_j^s, k_k^s will produce synchronized bursts as long as they are larger than the threshold k_s^* .

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